

Sam Shelly

Weed, CA | (530) 340-1065 | samgshelly@gmail.com | linkedin.com/in/sam-g-shelly | github.com/SamZombie | samzombie.github.io

EDUCATION

Vassar College

Aug 2022 – May 2026

Bachelor of Arts in Computer Science, Minor in Astronomy

Poughkeepsie, NY

GPA: 3.64/4.00 | Major GPA: 3.68/4.00

Relevant Coursework: Natural Language Processing, Numerical Methods, Analysis of Algorithms, Compilers, Theory of Computation, Data Structures & Algorithms

EXPERIENCE

AI Development Intern

May 2025 – Jan 2026

Innovative Computing Systems

Los Angeles, CA

- Hired directly by the COO to lead a Microsoft Copilot Studio Knowledge Agent pilot at a ~60-person company; worked full-time (30 hrs/week) as the primary builder on a 4-person team, meeting weekly with the COO to scope use cases.
- Built Python-based ingestion pipelines to index SharePoint content – including SOPs, the employee handbook, and company org hierarchy – for retrieval, and implemented an API integration connecting the agent to the company’s work-ticket system, letting engineers instantly query prior work history on a given client 30% faster lookups.
- Rolled the agent out to a daily-use focus group of 15 users and began scoping it for a second prospective external client as part of early efforts to commercialize the tool.

Computer Science Dept. Coach

Jan 2024 – May 2026

Vassar College

Poughkeepsie, NY

- Selected for a formal, paid CS Dept Coach position (8 hrs/week), retained across 4 consecutive semesters spanning two course iterations under two different instructors.
- Led 3 weekly sessions on algorithms, data structures, and debugging in Java for up to 50 students per semester (200+ total), adapting pacing and session plans for struggling students.

PROJECTS

AI Knowledge Base RAG

GitHub

Python, FastAPI, LangChain, Qdrant, Ollama, Mistral 7B, BAAI/bge-large-en-v1.5, PyMuPDF, Docker, Streamlit, pytest, GitHub Actions

Closed-loop RAG system built entirely on locally-hosted, open-weight components (Mistral 7B via Ollama, Qdrant) with no external API calls – suited to air-gapped and other network-restricted deployments. Applied to query government/defense AI policy documents, eliminating cloud dependency and controlling data exposure for secure, privacy-sensitive environments.

- By building a custom LLM-as-judge evaluation pipeline, achieved 0.95 retrieval recall and 0.92 answer similarity across 10 ground-truth QA pairs, visualized in a Streamlit dashboard.
- Ingests PDFs via PyMuPDF with overlapping chunking; embeds via BAAI/bge-large-en-v1.5 into Qdrant.
- Retrieves top-k chunks and generates grounded, cited answers via Mistral 7B (Ollama).
- Served via FastAPI REST API, containerized with Docker Compose, with CI/CD via GitHub Actions.

PUBLICATIONS

Feedback-Driven Magnetic Field Evolution in MW-type Simulated Circumgalactic Medium

2024

Param P. Gogia, Umman Azan, **Sam Shelly**, Edward Buie II | *KNAC Undergraduate Symposium Proceedings*

Investigated turbulence-driven feedback’s role in magnetic field evolution in the CGM of a simulated Milky Way-type galaxy across three MAIHEM MHD simulations (two solar-metallicity, one at 0.3 $Z_{\odot}$). Found that feedback induces late-stage magnetic field saturation via disk instability and outward propagation of cold, magnetized gas, versus steady cubic growth in the no-feedback case.

My contribution: performed the post-simulation analysis (Mach-Alfvén profiles, angular momentum phase plots, magnetic/kinetic energy density evolution), built all data visualizations, and wrote ~50% of the manuscript.

Efficacy of NER Models on Fiction and Nonfiction Domains

2025

Duncan Beauchamp, **Sam Shelly** | *CMPU 366 Natural Language Processing, Vassar College*

Evaluated cross-domain performance of a DistilBERT-based NER model on literary fiction (LitBank) vs. expository text (OntoNotes), restricted to shared PER/LOC/ORG entity types. A pretrained baseline showed a 0.368 F1 domain gap between genres; fine-tuning on a mixed-domain corpus (3× oversampled LitBank, 50-token chunking, literary-only validation split) cut the gap to 0.128, raising LitBank micro-F1 from 0.26 to 0.72.

My contribution: built the training data pipeline (dataset loading, chunking, oversampling, train/validation/test splits) and wrote ~50% of the manuscript.

SKILLS

Languages: Python, Java, C, C++, OCaml | **ML/AI & Data:** PyTorch, HuggingFace Transformers, DistilBERT, scikit-learn, LangChain, Ollama, Qdrant, pandas, yt-project | **Tools:** Docker, FastAPI, GitHub Actions, PyMuPDF